DSA 210 Introduction to Data Science

2025–2026 Spring Term

FINAL PROJECT REPORT

Can Weather Predict Water? Analyzing Istanbul Dam Occupancy Through Meteorological Data

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GitHub Repository: github.com/aysenuruyar-rgb/DSA210-Proj

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1. Motivation

Water sustainability is one of the most pressing challenges facing rapidly growing metropolitan cities. Istanbul, home to over 15 million residents, depends almost entirely on a network of reservoirs and dams for its freshwater supply. The city has experienced water shortages in the past, and with the dual pressures of population growth and climate change intensifying, understanding what drives reservoir levels is not merely an academic exercise — it is a public health and urban planning priority.

This project was motivated by a deceptively simple question: can daily weather — specifically rainfall and snowfall — meaningfully explain or predict how full Istanbul's dams are on any given day? The intuitive answer is yes: it rains, water levels rise. But water systems are physically complex. Precipitation must saturate the soil, overcome evaporation, run off into catchment areas, and accumulate over time before reaching a reservoir. Daily weather data may therefore be a very noisy proxy for the slow-moving dynamics that actually govern dam levels.

By rigorously testing this intuition with real historical data, this project aims to reveal where the predictive signal truly lies — and what that means for how we should model and manage urban water resources.

2. Data Sources

Two complementary datasets were collected, merged, and aligned by date to form the analytical foundation of this project. Together they span from the early 2000s to the present day, providing over two decades of daily observations.

2.1 Dam Occupancy Data

- Source: Istanbul Metropolitan Municipality Open Data Portal (IBB Açık Veri Portali)
- Granularity: Daily
- Key feature: Reservoir occupancy rate (%), representing the aggregate fill level of Istanbul's dams
- Collection method: Direct download from the official open data platform

This dataset is authoritative and publicly maintained by the municipal authority responsible for Istanbul's water infrastructure, making it highly reliable. The occupancy rate is a summary statistic that aggregates across multiple reservoirs in the Istanbul water supply network.

2.2 Weather Data

- Source: Open-Meteo Historical Weather API
- Granularity: Daily

- Key features: Rainfall (mm), Snowfall (cm), Air Temperature (°C)
- Collection method: Automated Python API requests, one record per day for Istanbul coordinates

Open-Meteo provides reanalysis-based historical weather data and is widely used in research contexts. Querying it programmatically using Python ensured reproducibility. The three weather variables selected represent the primary meteorological inputs into any surface hydrology system.

2.3 Data Integration

Both datasets were aligned on a common date index and merged into a single master file (final_dataset.csv). Missing values were identified and handled during the data preparation stage. A second enriched file (featured_dataset.csv) was subsequently created after feature engineering.

3. Data Analysis

The analysis followed a structured five-stage pipeline, progressing from raw data to actionable insights. Each stage is documented in a dedicated Jupyter notebook.

3.1 Data Preparation (notebook/01_data_preparation.ipynb)

The first stage focused on data quality and integration. The dam occupancy and weather datasets were loaded separately, aligned on a shared daily date index, and merged into a single clean dataframe. Missing values were examined and handled appropriately. The resulting clean dataset was saved as data/processed/final_dataset.csv and served as the starting point for all subsequent analysis.

3.2 Exploratory Data Analysis (notebook/02_eda.ipynb)

EDA was used to characterise the data and identify initial patterns before any formal testing. The following visualisations and analyses were conducted:

- Distribution analysis via histograms for dam occupancy, rainfall, and snowfall
- Time series plots to observe long-run trends and year-to-year variation in dam levels
- Scatter plots examining the pairwise relationship between each weather variable and dam occupancy
- Seasonal decomposition and monthly comparisons to explore intra-year cycles
- Correlation matrix to summarise linear relationships between all variables

Key EDA observations: dam occupancy showed clear seasonal oscillation with peaks in spring and troughs in late summer/autumn. Rainfall and snowfall showed weak unconditional correlation with dam levels, consistent with the hypothesis that the relationship is lagged or non-linear rather than contemporaneous.

Figure 1 below visualizes the time-series relationship between dam occupancy and rainfall. The plot shows that dam occupancy follows smoother seasonal cycles, while rainfall fluctuates more sharply, supporting the idea that reservoir levels do not respond instantly to same-day precipitation.

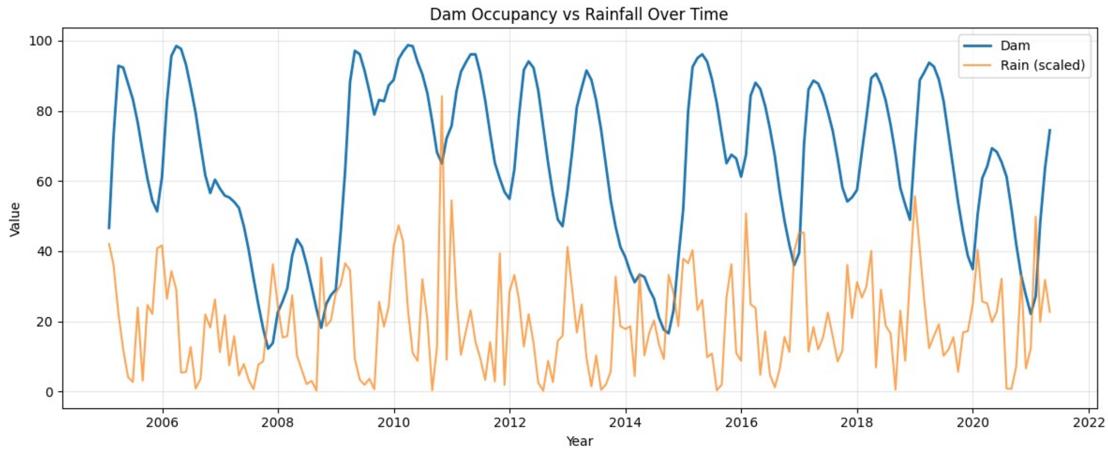

Figure 1. Dam occupancy and scaled rainfall over time.

The correlation matrix further confirms that the direct linear relationship between daily weather variables and dam occupancy is weak.

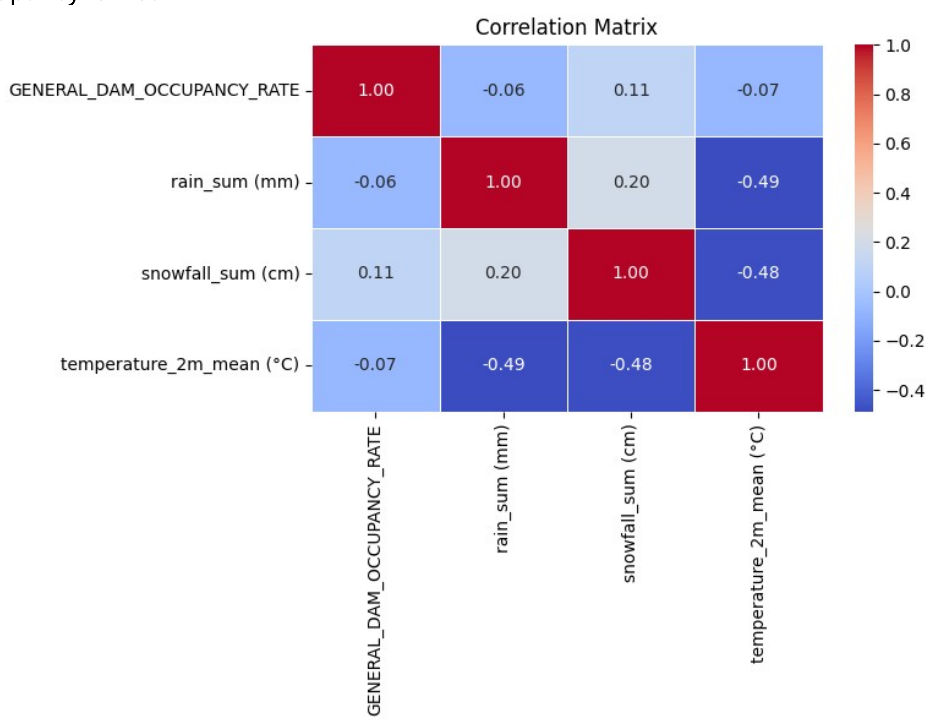

Figure 2. Correlation matrix of dam occupancy and weather variables.

3.3 Hypothesis Testing (notebook/03_hypothesis_tests.ipynb)

Three formal statistical hypotheses were evaluated:

Hypothesis	Description	Test Used	Result
H1	Daily rainfall significantly affects dam occupancy	Pearson correlation / t-test	Not significant ($p \approx 0.415$)
H2	Weather variables (rain, snow, temp) collectively explain dam levels	Multiple linear regression (OLS)	Very weak ($R^2 \approx 0.02$)
H3	Seasonality (month/quarter) significantly affects dam occupancy	ANOVA / Kruskal-Wallis	Confirmed ($p < 0.001$)

H1 finding: with a p-value of approximately 0.415, there is no statistically significant linear relationship between the amount of rainfall on a given day and the dam occupancy level on that same day. This does not mean rain is irrelevant — it means the effect is not visible at the daily contemporaneous scale.

H2 finding: a multiple regression using rainfall, snowfall, and temperature as predictors explained only around 2% of the variance in dam occupancy. This confirms that short-term weather variables are individually and jointly insufficient to model reservoir dynamics.

H3 finding: dam occupancy levels differed significantly across seasons and months ($p < 0.001$). This was the strongest statistical result in the study and points to seasonality — capturing cumulative precipitation patterns and evaporation cycles — as the dominant structural driver.

The seasonal pattern is also visible in the average occupancy by season: spring has the highest mean occupancy, while fall has the lowest. This supports the ANOVA result and highlights the importance of seasonal accumulation.

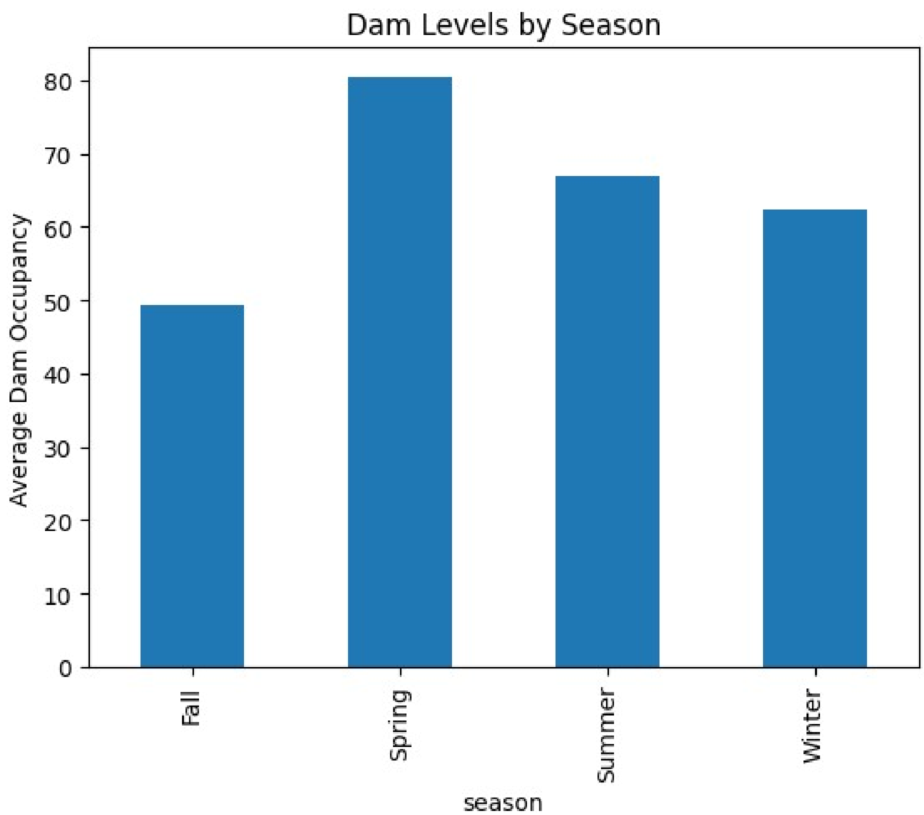

Figure 3. Average dam occupancy by season.

3.4 Feature Engineering (notebook/04_feature_engineering.ipynb)

Given that contemporaneous weather variables alone were weak predictors, additional features were engineered to capture the temporal memory of the system:

- Lag variables: previous dam occupancy levels (e.g., 1-day, 7-day, 30-day lags) and lagged weather values to capture delayed effects of precipitation
- Rolling averages: 3-month (90-day) moving averages of rainfall, snowfall, and dam occupancy to smooth short-term noise and capture accumulated trends
- Combined precipitation: a unified precipitation variable merging rainfall (mm) and snowfall (converted to mm water equivalent) into a single total precipitation metric
- Seasonal encoding: sine and cosine transformations of the day-of-year variable to encode the cyclical nature of seasons in a continuous, model-friendly format
- Binary rain indicator: a boolean is_rainy flag for days with measurable precipitation, allowing models to distinguish wet from dry days categorically

The enriched dataset was saved as data/processed/featured_dataset.csv and used as input for machine learning modelling.

3.5 Machine Learning Modelling (notebook/05_modeling.ipynb)

Two regression models were trained to predict daily dam occupancy using the engineered feature set:

- Linear Regression: a parametric baseline model that assumes linear relationships between features and the target
- Random Forest Regressor: a non-parametric ensemble model capable of capturing non-linear interactions and feature importance

Both models were evaluated using Mean Absolute Error (MAE) and R-squared (R^2). Results showed that adding lag variables and seasonal encodings dramatically improved linear regression performance compared to the baseline model trained on raw weather variables alone. The Random Forest further improved upon this by capturing non-linear interactions, particularly between lagged dam levels and seasonal features.

The final model performance values are summarized below:

Model	Features	Metric	R^2
Linear Regression	Weather + lag + rolling + seasonal features	MAE = 2.99	0.955
Random Forest Regressor	All engineered features	MSE = 25.18	0.937

These results show that engineered temporal features substantially improve prediction performance. The high R^2 values indicate that historical occupancy patterns, rolling averages, and seasonal structure are much stronger predictors than short-term weather variables alone.

The most important predictors across both models were temporal features — specifically lagged dam occupancy values and seasonal encodings. Current-day weather variables ranked low in feature importance, reinforcing the hypothesis testing conclusions.

4. Key Findings

The analysis produced four central findings that together tell a coherent story about Istanbul's water system:

Finding 1 — Daily Rainfall Does Not Predict Dam Levels

Despite the intuitive expectation that rain raises reservoir levels, no statistically significant relationship was found between daily rainfall and same-day dam occupancy ($p \approx 0.415$). Istanbul's catchment hydrology, urban runoff dynamics, and the sheer size of the reservoir network mean that any single day's rain is effectively invisible in the aggregate occupancy statistic.

Finding 2 — Short-Term Weather Has Very Low Explanatory Power

All three weather variables combined (rainfall, snowfall, temperature) explain only about 2% of the variance in dam occupancy ($R^2 \approx 0.02$). This is a striking result: 98% of the variation in reservoir levels cannot be accounted for by looking at today's weather. Water systems have long memories — what matters is accumulated precipitation over weeks and months, not what fell today.

Finding 3 — Seasonality Is the Dominant Driver

The strongest statistical signal in the entire analysis was seasonality. Dam occupancy varies significantly by month and season ($p < 0.001$), with a consistent pattern: levels are highest in spring following winter and early-spring precipitation accumulation, and lowest in late summer/early autumn after the hot dry season. This seasonal rhythm is a far better predictor than any individual weather variable.

The monthly heatmap provides an additional view of this pattern across years, showing repeated spring peaks and late-year declines.

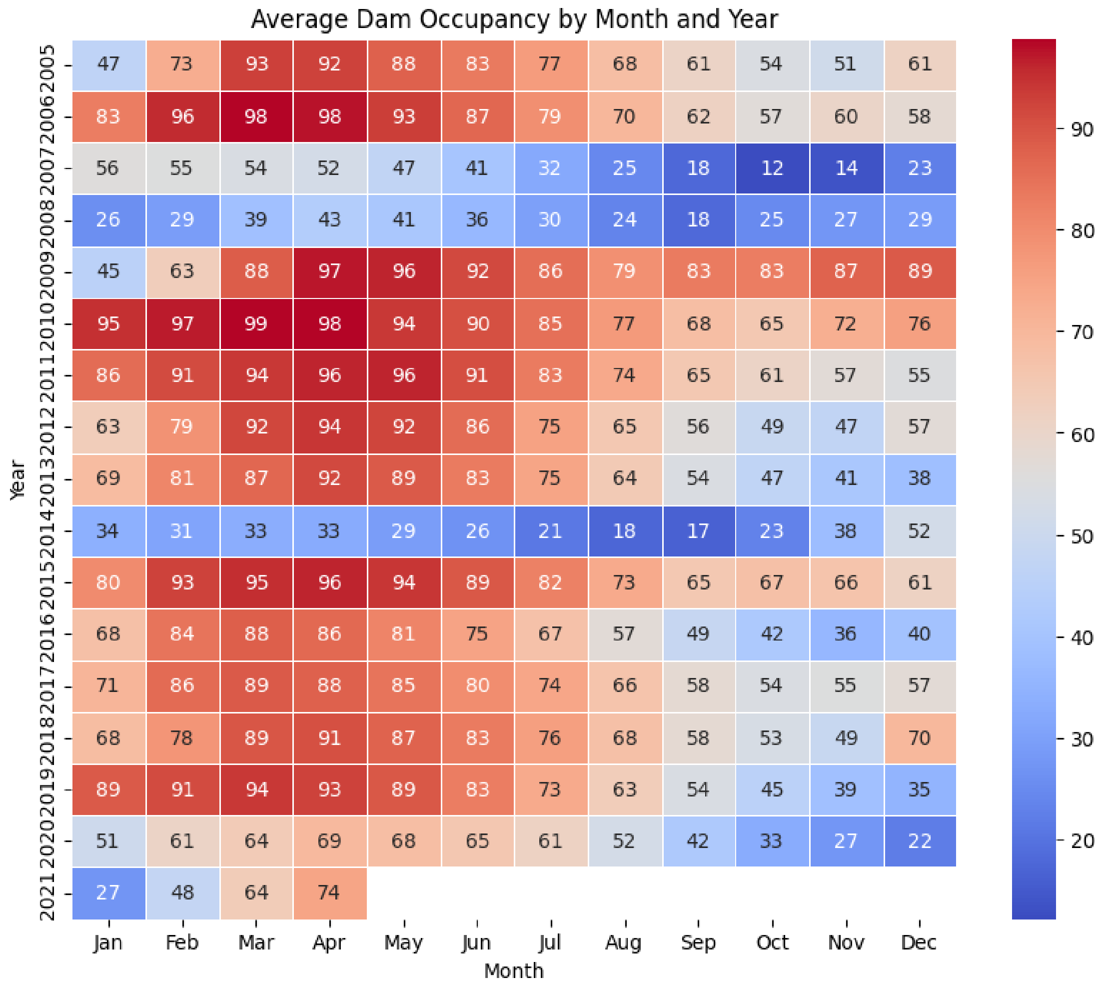

Figure 4. Average dam occupancy by month and year.

Finding 4 — Predictive Models Improve with Temporal Features

Machine learning models that incorporated lag variables (past dam levels) and seasonal encodings achieved substantially better predictive accuracy than those using only raw weather inputs. This confirms that dam occupancy is fundamentally an autoregressive time series: the best predictor of today's reservoir level is yesterday's reservoir level, modulated by seasonal patterns accumulated over time.

5. Limitations and Future Work

5.1 Current Limitations

- Single feature for dam occupancy: the target variable is an aggregate percentage across multiple reservoirs. Individual dams likely behave differently due to location, catchment area, and management rules. Disaggregated data would enable richer modelling.
- No consumption or withdrawal data: dam occupancy is a net quantity — it reflects both inflows (precipitation-driven) and outflows (human consumption, evaporation, controlled releases). Without controlling for water demand, any weather-level model is misspecified by construction.
- Temporal granularity mismatch: daily weather data was used to explain a daily reservoir metric, but the physical processes (soil saturation, runoff, subsurface flow) operate on time scales of days to weeks. The lag analysis partially addresses this but does not fully capture the hydrological transfer function.
- No non-linear precipitation effects: the hypothesis tests used linear methods. Extreme rainfall events (e.g., storms) may have disproportionate effects that are invisible to Pearson correlation.
- Limited geographic weather data: a single-point weather observation was used for the entire Istanbul region. Istanbul spans a large geographic area with varied topography; catchment-specific weather data would improve accuracy.

5.2 Future Extensions

- Incorporate population growth and per-capita water consumption data to model net dam inflows rather than gross occupancy.
- Apply time series models (ARIMA, SARIMA, Prophet) that are specifically designed for autoregressive seasonal data.
- Use spatial weather data (gridded reanalysis products) matched to each dam's catchment area rather than a single regional weather station.
- Investigate the impact of individual extreme weather events (storms, drought periods) on reservoir recovery time.
- Expand the study to compare Istanbul with other Turkish cities or Mediterranean cities facing similar water stress.
- Incorporate snowpack data where available, since winter snow accumulation and spring melt may drive a significant share of annual reservoir recharge.

6. AI Disclosure

In accordance with course guidelines on academic integrity, the use of AI tools is disclosed here explicitly. ChatGPT (OpenAI) was used during this project for the following purposes:

- Improving documentation clarity and phrasing in the README and report sections
- Refining written explanations of statistical concepts for a general audience
- Debugging Python code and explaining error messages

All data collection, analytical decisions, statistical interpretation, feature engineering choices, and machine learning implementation were conducted and verified independently by the author. AI assistance was limited to language and code explanation tasks and did not substitute for original analytical work.

7. Conclusion

This project set out to answer whether daily weather conditions are sufficient to explain Istanbul's dam occupancy levels. The answer is clearly no — but the reasons why are instructive. Water systems are not instantaneous: they accumulate, retain, and release water over timescales much longer than a single day. The dominant predictor of a reservoir's fill level on any given day is where it was yesterday, shaped by the slow seasonal rhythm of a Mediterranean climate.

These findings have practical implications. Water resource managers should not rely on short-term weather forecasts to anticipate reservoir dynamics. Instead, long-range seasonal outlooks, multi-week precipitation accumulations, and demand-side modelling are needed for effective planning. From a data science perspective, this project demonstrates that domain understanding — knowing that hydrological systems have memory and seasonality — is essential for choosing the right features and interpreting results correctly.

The project followed a complete data science pipeline: data collection from two independent sources, cleaning and merging, exploratory analysis, formal hypothesis testing, feature engineering, and supervised machine learning. Each stage built on the previous one and is fully reproducible via the linked GitHub repository.